404: Hallucination Not Found

RAG-Based Clinical Decision Support for Mental Health During Pregnancy
Guideline Source: NICE CG192

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1. Project Overview

1.1 Problem Statement

Standard large language models (LLMs), when applied to clinical settings, present three critical failure modes that make them unsafe for medical use:s

- **Frequent Hallucination** — models fill knowledge gaps with plausible-sounding but fabricated information.
- **Fabricated Clinical Advice** — invented citations, clinical trials, or incorrect medical guidelines presented as fact.
- **Dangerous False Confidence** — incorrect advice delivered with a convincing, authoritative tone, leaving clinicians unable to detect errors.

In the domain of perinatal mental health — where decisions affect both mother and unborn child — the stakes of a hallucinated drug dosage or missed contraindication are life-threatening. There is zero tolerance for inaccuracy.

1.2 Proposed Solution

404: Hallucination Not Found is a Retrieval-Augmented Generation (RAG) system that grounds every clinical answer directly in the **NICE CG192** guideline — the official UK clinical guideline for antenatal and postnatal mental health. The architecture guarantees:

- Every answer is traceable to a specific chunk of the verified guideline.
- Out-of-scope or low-confidence queries are refused gracefully rather than hallucinated.
- Clinicians can inspect the exact retrieved evidence supporting each response.

1.3 Scope & Clinical Context

The system is strictly scoped to NICE CG192 and handles high-stakes clinical scenarios including:

- Pharmacological management during pregnancy (e.g., lithium, SSRIs, TCAs, SNRIs).
- Diagnosis and screening procedures for conditions such as bipolar disorder in pregnant women.
- Management of severe presentations such as tokophobia (extreme fear of childbirth).
- Safety monitoring and contraindication checks for perinatal drug regimens.

⚕ **Clinical Boundary:** The system does NOT answer questions outside the NICE CG192 guideline. When no matching evidence is found (similarity < 0.70), the system issues a graceful refusal and redirects the clinician to specialist documentation.

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2. System Architecture & RAG Pipeline

The system follows a classic Retrieval-Augmented Generation architecture with five core stages:

2.1 Stage 1 — Document Parsing

Tool: LlamaParse
Output: parsed_document.json

The NICE CG192 PDF is ingested using LlamaParse, which extracts structured text while preserving section boundaries, headers, and page context. This structured output forms the raw input for all downstream processing.

2.2 Stage 2 — Text Cleaning

Tool: Custom Regex noise-stripper
Output: cleaned_document.json

A regex-based cleaning pipeline removes artifacts introduced during PDF extraction: excessive whitespace, hyphenation remnants, header/footer repetition, and non-informative characters. This improves embedding quality by ensuring chunks contain meaningful clinical content.

2.3 Stage 3 — Text Chunking

Tool: LangChain RecursiveCharacterTextSplitter
Output: chunks.json

Parameter	Value	Rationale
Chunk size	300 tokens	Balances specificity and context length
Overlap	100 tokens	Preserves cross-boundary clinical context
Top-k retrieval	5 chunks	Filters noise while covering relevant sections
Splitter type	Recursive	Respects natural language boundaries

Each chunk includes metadata: page number, section heading, source document, and character count — enabling precise citation in responses.

2.4 Stage 4 — Embedding Generation

Model: sentence-transformers/all-MiniLM-L6-v2
Output: 384-dimensional float vector per chunk

Each cleaned chunk is converted into a dense numerical vector using the all-MiniLM-L6-v2 SentenceTransformer model. This same model is used at query time — the user's clinical question is embedded identically, enabling semantic (meaning-based) similarity comparison rather than keyword matching.

2.5 Stage 5 — Vector Database Storage

Database: ChromaDB PersistentClient
Collection: nice_cg192_mental_health

ChromaDB stores all chunk embeddings, the original chunk text, and per-chunk metadata in a persistent local collection. At query time, the user's query embedding is matched against stored chunk embeddings using cosine similarity, and the top-5 most similar chunks are returned.

Each stored record contains:

- **id** — unique chunk identifier
- **document** — the raw chunk text
- **embedding** — 384-dim float vector

- **metadata** — page, section, source, char_count

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3. Retrieval & Safety Mechanisms

3.1 Semantic Similarity Retrieval

When a clinician submits a query, the following sequence runs:

- Query is embedded using all-MiniLM-L6-v2 into a 384-dim vector.
- ChromaDB performs cosine similarity scoring against all stored chunk vectors.
- The top-5 scoring chunks are returned as candidate evidence.
- Each result includes a similarity score, chunk text, and metadata.

Cosine similarity is chosen for its ability to measure directional alignment between vectors independently of magnitude — ideal for short clinical text snippets.

3.2 Clinical Safety Threshold & Refusal

Similarity Threshold: ≥ 0.70 → Return grounded evidence + citation
Below Threshold: < 0.70 → Graceful refusal — Out of Guideline Scope

This threshold is the core safety guardrail. If no retrieved chunk achieves a similarity score of 0.70 or higher, the system does not generate an answer. Instead, it:

- States that the query falls outside the indexed guideline scope.
- Refuses to hallucinate or speculate.
- Redirects the clinician to consult a human specialist or the primary NICE CG192 documentation.

3.3 Refusal Quality Rubric

The system scores **3/3** on the internal refusal quality rubric:

Criterion	Description	Status
States Insufficiency	Clearly detects and communicates missing or incomplete guideline context.	✓ Pass
Stays Honest	Refuses to invent citations, hallucinate advice, or offer speculative guidance.	✓ Pass
Offers Next Step	Safely redirects to human specialists or official primary documentation.	✓ Pass

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4. Grounded Response Generation

4.1 LLM Generation with Groq API

Once high-similarity evidence chunks are retrieved, they are packaged into a structured prompt and passed to the Groq API for generation. The system prompt explicitly instructs the model to:

- Answer exclusively using the retrieved guideline evidence provided in context.
- Cite the specific chunk IDs from which each claim is drawn.
- Not supplement with general training knowledge outside the provided context.
- If context is insufficient, refuse and state why.

This prompt engineering approach is the second layer of hallucination prevention, working in concert with the retrieval threshold.

4.2 Citation & Evidence Traceability

Every response surfaces the exact evidence behind the answer. The flow is:

- User query → Similarity search → Extracted guideline chunks
- Chunks → In-text citation by chunk ID → Direct source reference
- Source reference → Linkable passage from NICE CG192

This traceability allows any clinician to verify the AI's answer against the authoritative guideline in a single step.

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5. Evaluation Metrics

5.1 Metrics Defined

- **# Retrieved Chunks** — number of chunks returned above the similarity threshold.
- **Precision@5** — proportion of the top-5 retrieved chunks that are genuinely relevant.
- **Citation Accuracy** — whether the response correctly cited the source chunks (0 or 1).
- **Faithfulness** — proportion of response claims grounded in the retrieved chunks (0–1 scale).

5.2 Evaluation Results

Prompt	Chunks	Precision@5	Citation Acc.	Faithfulness
Sertraline at 20 weeks — should I stop? (out-of-scope)	0	0	0	0
How is bipolar diagnosed for pregnant women?	4	0.80	1	0.833

Prompt	Chunks	Precision@5	Citation Acc.	Faithfulness
Managing tokophobia / extreme fear of childbirth	1	0.20	1	0.500
Treatment options: TCAs, SSRIs, or SNRIs?	3	0.60	1	0.667
Safety risks & guidelines for lithium during pregnancy	5	1.00	1	0.750

5.3 Key Findings

- The out-of-scope patient query (sertraline) was correctly refused with all metrics at 0 — the safety threshold working as intended.
- The lithium query achieved perfect Precision@5 and Citation Accuracy with strong faithfulness (0.75), confirming reliable coverage of the most complex pharmacological content.
- The tokophobia query had low Precision@5 (0.20) indicating sparse coverage in the guideline, but the system cited accurately and remained faithful to retrieved context.
- Citation Accuracy was **1.0 for all in-scope queries** — every answer correctly traced its claims back to a guideline chunk.

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6. Technology Stack

Component	Technology	Role
Frontend / UI	Streamlit	Web interface for clinical queries and evidence display
Backend language	Python	Main application programming language
Embedding model	SentenceTransformers all-MiniLM-L6-v2	Converts text to 384-dim semantic vectors
Vector database	ChromaDB	Stores and retrieves embeddings via cosine similarity
LLM provider	Groq API	Generates grounded clinical responses
Document parser	LlamaParse	Extracts structured text from the NICE CG192 PDF
Chunker	LangChain RecursiveCharacterTextSplitter	Splits text into overlapping semantic chunks
UI customisation	HTML / CSS	Streamlit interface theming and layout
Version control	Git & GitHub	Source control and project repository
Deployment	Streamlit Community Cloud	Hosts the live web application
Secret management	Streamlit Secrets	Secures the Groq API key in production

7. Deployment Workflow

7.1 Local Development & Testing

The full RAG pipeline — parsing, cleaning, chunking, embedding, ChromaDB ingestion, and Streamlit UI — was built and validated locally in Python. Test queries were run against the local ChromaDB collection to verify retrieval quality, threshold behaviour, and response faithfulness before any deployment step.

7.2 Version Control

The project is tracked with Git and hosted on GitHub. A `requirements.txt` file captures all Python dependencies to ensure environment reproducibility. A `.gitignore` file prevents sensitive files (API keys, local ChromaDB files) from being committed to the repository.

7.3 Streamlit Cloud Deployment

The GitHub repository is linked to Streamlit Community Cloud, which automatically:

- Reads `requirements.txt` and installs all dependencies.
- Runs `app.py` as the live web application entry point.
- Rebuilds and redeploys on every push to the main branch.

7.4 Secret Management

The Groq API key is never hardcoded or committed. It is configured through Streamlit Secrets and accessed at runtime via `st.secrets["GROQ_API_KEY"]`. This ensures the production key is never exposed in the codebase.

7.5 Cloud vs Local Environment Handling

A locally generated ChromaDB collection cannot be assumed to exist in the cloud environment. The guideline data and vector store are configured to be available to the deployed application, and the retrieval layer is initialised accordingly on first run in the cloud.

Full pipeline: Local Development → Git & GitHub → Streamlit Cloud → Configure Secrets → Live Web App

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8. Limitations & Future Work

8.1 Current Limitations

- Scope is strictly limited to NICE CG192 — no cross-guideline retrieval.
- Tokophobia coverage is sparse in the guideline, resulting in low Precision@5 for that query type.
- The similarity threshold (0.70) was manually calibrated — an adaptive threshold tuned per query type could improve precision.
- The system produces text responses only; clinical decision trees or structured treatment pathways are not yet generated.
- No user authentication layer — the current deployment is open access.

8.2 Proposed Extensions

- **Multi-guideline RAG** — extend the vector store to include NICE CG192 alongside related guidelines (e.g., perinatal pharmacovigilance databases).
- **Adaptive thresholding** — use query classification to apply different similarity thresholds for different clinical question types.
- **Automated evaluation pipeline** — integrate RAGAS or a similar RAG evaluation framework for continuous quality monitoring.
- **Structured output formats** — generate summary tables, dosage charts, or contraindication matrices alongside free-text answers.
- **Clinician feedback loop** — capture clinician ratings of response usefulness to fine-tune retrieval ranking over time.

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9. Glossary

Term	Definition
RAG	Retrieval-Augmented Generation — an LLM architecture that grounds responses in retrieved documents rather than relying solely on training data.
NICE CG192	National Institute for Health and Care Excellence Clinical Guideline 192 — the UK's official guideline on antenatal and postnatal mental health.
Hallucination	When an LLM generates plausible-sounding but factually incorrect or fabricated information.
Embedding	A dense numerical vector representation of text that captures semantic meaning.
ChromaDB	An open-source vector database used to store and query embeddings.
Cosine Similarity	A metric measuring the angular similarity between two vectors — used to rank retrieved chunks by relevance.
Precision@5	The proportion of the top-5 retrieved items that are genuinely relevant to the query.
Faithfulness	The proportion of a generated response's claims that are directly supported by the retrieved evidence.

Term	Definition
Similarity Threshold	A minimum similarity score (0.70) below which the system refuses to generate a response.
Groq API	A high-speed LLM inference API used to generate the final natural language response.
SentenceTransformers	A Python library for computing dense sentence and text embeddings using transformer models.
Streamlit	A Python framework for building interactive web applications, used here as the clinical UI.
LangChain	A framework for building LLM-powered applications, used here for text chunking.

Halting AI Hallucination, Healing Trust.